Chapter 4: How Plants Make Their Food

Every living organism requires food. Food gives us energy to move, grow and repair our bodies. Animals, including humans, obtain food by eating plants or other animals. Green plants are different. They are able to manufacture their own food, and they utilize nothing more than sunlight, water and air to do it. In this chapter we will find out how plants accomplish this remarkable task.

4.1 What is Photosynthesis?

Photosynthesis is the process by which green plants make their own food using sunlight, water and carbon dioxide. The word comes from two Greek words: photo, meaning light, and synthesis, meaning putting together. During photosynthesis, plants put together simple substances to form a sugar called glucose, and oxygen is released into the air as a by-product.

We can write photosynthesis as a simple word equation: carbon dioxide plus water, in the presence of sunlight and chlorophyll, gives glucose plus oxygen. Organisms that make their own food in this way are called autotrophs. Organisms that depend on others for food, such as animals and fungi, are called heterotrophs.

4.2 The Leaf: A Food Factory

Although every green part of a plant can carry out photosynthesis, leaves are the main food factories. A leaf is broad and flat, so it can absorb as much sunlight as possible. Numerous veins run through the leaf and transport water and minerals to every cell.

If you look at the underside of a leaf through a microscope, you will observe tiny pores on its surface. These pores are called stomata. Each pore is surrounded by a pair of guard cells, which open and close the pore. Carbon dioxide from the air enters the leaf through the stomata, and oxygen produced during photosynthesis leaves through them.

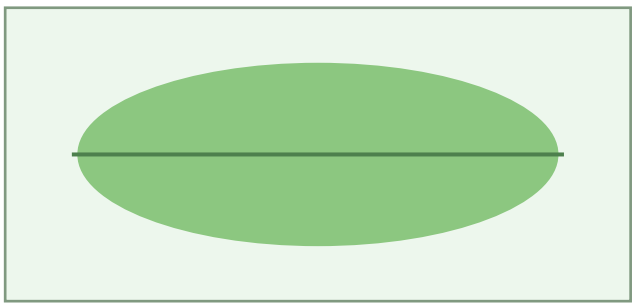

Figure 4.1: A cross-section of a leaf showing stomata, guard cells and chloroplasts.

4.3 The Role of Chlorophyll

Chlorophyll is a green pigment found in the leaves of plants. It is present inside tiny structures called chloroplasts. Chlorophyll captures the energy of sunlight, and this energy is subsequently used to split water and combine it with carbon dioxide. Without chlorophyll, photosynthesis cannot take place. This is why leaves that are not green, such as the white parts of a variegated leaf, do not produce starch.

4.4 Water and Transpiration

Plants absorb water from the soil through their roots. Root hairs increase the surface area of the roots, so they can take in more water and minerals. The water travels up through the stem in thin tubes called xylem and finally reaches the leaves.

Plants lose a large amount of water as vapour through their stomata. This loss of water from the leaves is called transpiration. Transpiration is not wasteful; it creates a pull that draws water upward from the roots, and it also cools the plant on hot days, much like sweating cools our bodies.

4.5 What Happens to the Food?

The glucose made in the leaves is used by the plant for energy. Extra glucose is converted into starch and stored in different parts of the plant, such as roots, stems, fruits and seeds. Potatoes, rice and wheat are rich in starch because the plant stores its food there. We can test a leaf for starch in the laboratory.

- Take a leaf from a plant that has been kept in sunlight for several hours.
- Boil the leaf in water for one minute, then boil it in alcohol to remove the chlorophyll.
- Rinse the leaf in warm water and spread it on a dish.
- Add a few drops of iodine solution. A blue-black colour shows that starch is present.

4.6 Why Photosynthesis Matters

Photosynthesis is the foundation of almost every food chain on Earth. When a deer eats grass, or a tiger eats a deer, the energy they receive was originally captured from sunlight by a plant. Photosynthesis also maintains the level of oxygen in the atmosphere, which nearly all living organisms require for respiration. Consequently, protecting forests and green spaces is essential for life on our planet.

In summary, green plants use chlorophyll to capture sunlight, take in carbon dioxide through their stomata and water through their roots, and make glucose and oxygen. Some of the glucose is used immediately and the rest is stored as starch for later.